

QCI Central Finite Curve

Production Implementation Plan

Prepared: 24 April 2026 · Post-Demo with Subroto Sir (23 Apr 2026)

Status: Decision Document for Leadership

Executive Summary

The demo successfully validated the core concept: a RAG-powered platform that ingests QCI's institutional knowledge (Work Orders, Proposals) and generates new drafts grounded in prior work. The current system runs on Railway (PaaS) with a Next.js frontend, FastAPI backend, PostgreSQL + pgvector, Redis/Celery, RunPulse OCR, and Gemini Flash for generation and embeddings.

This document presents three implementation options — from lean to enterprise-grade — so leadership can choose based on budget appetite, risk tolerance, and timeline.

⚠ IMPORTANT

The demo processed ~24 documents. Production requires ingesting ~80 GB of institutional documents and supporting ~500 concurrent users. This is a fundamentally different scale requiring deliberate architecture decisions.

1. Current Architecture — Demo State

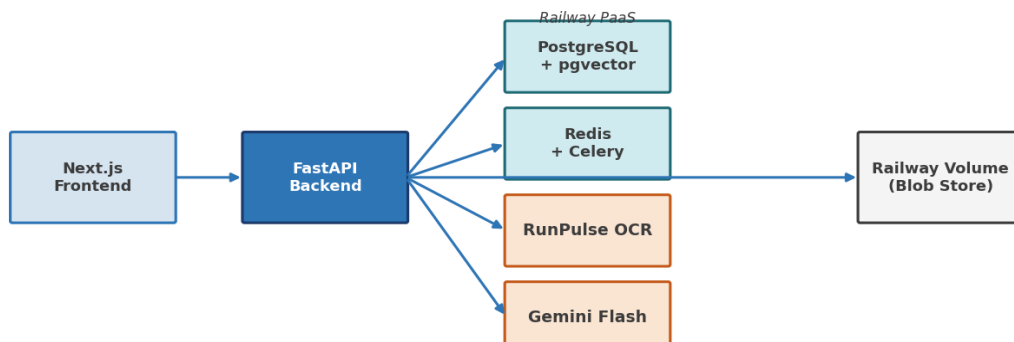


Figure 1: Demo system components running on Railway PaaS

Component	Demo Setup	Production Gap
Documents	24 PDFs (~50 MB)	~80 GB corpus (thousands of docs)
Users	1–3 demo users	~500 concurrent
Auth	Shared demo password + JWT	SSO / RBAC needed
OCR	RunPulse free/dev tier	Enterprise subscription required
Database	Railway-managed Postgres	Needs backup, HA, proper sizing
Storage	Railway volume mount	S3-compatible blob storage
LLM	Gemini Flash free tier	Paid tier or alternative needed

2. Key Decision Areas

2.1 OCR — Processing 80 GB of Documents

The current pipeline uses RunPulse (Pulse AI) for OCR and schema extraction. RunPulse does not publish fixed pricing — enterprise is custom/sales-driven.

Estimation basis: 80 GB ≈ 15,000–25,000 document pages (assuming average 3–5 MB per multi-page PDF).

OCR Option	One-Time Ingest Cost	Ongoing Cost	Quality	Effort to Switch
RunPulse Enterprise	₹3–6L (custom quote: on the higher side ~10k pages are free)	₹50K–1L/mo	High (tables, layout-aware)	None — already integrated
Google Document AI	~₹3–5L for full corpus	Pay-per-page	High	Medium (API swap)
Docling (Open Source)	₹0 + compute time	Compute only	Good (improving rapidly)	Low — already have prior experience
Azure AI Doc Intelligence	~₹2–4L	Pay-per-page	High	Medium

✓ **RECOMMENDATION** Use Docling (open-source, with prior experience from the reimbursement project) for bulk historical ingestion to save ₹3–6L, and retain RunPulse only for ongoing new documents where structured schema extraction is critical. This hybrid approach minimises cost while maintaining quality. Can check with RunPulse Sales team to get a realistic quote for services.

2.2 Hosting & Infrastructure

The core question: can Railway handle 500 concurrent users, or do we need Azure/AWS?

Factor	Railway Pro/Enterprise	Azure App Service	AWS Fargate
500 concurrent users	Yes (horizontal replicas)	Yes (scale-out plans)	Yes (auto-scaling)
Ease of Setup	★★★★★ — git push deploy	★★★★☆ — more config	★★★☆☆ — DevOps heavy
Maintenance Burden	Very low	Medium	High
Est. Monthly Cost	₹8K–25K	₹15K–40K	₹12K–35K
India Region	No (US/EU only)	Yes — Central India	Yes — Mumbai
Compliance / Govt	Limited	Strong (govt certs)	Strong
Data Residency	US/EU servers	India possible	India possible

■ DATA RESIDENCY WARNING

Data residency is critical for a government body like QCI. Railway hosts in US/EU only. If QCI's IT security policy requires data to stay in India, Azure (Central India) or AWS (Mumbai) becomes mandatory regardless of convenience.

Azure vs AWS practical comparison for this project:

- Azure is the better fit if QCI already has Microsoft licensing (Office 365, etc.) — Azure Hybrid Benefit could cut costs 30–40%
- AWS offers more granular container control but requires more DevOps expertise
- Both provide equivalent compute for this use case; the difference is in operational complexity, not capability

2.3 Database & Deployment Architecture

Current: Railway-managed PostgreSQL (auto-provisioned, limited backup control).

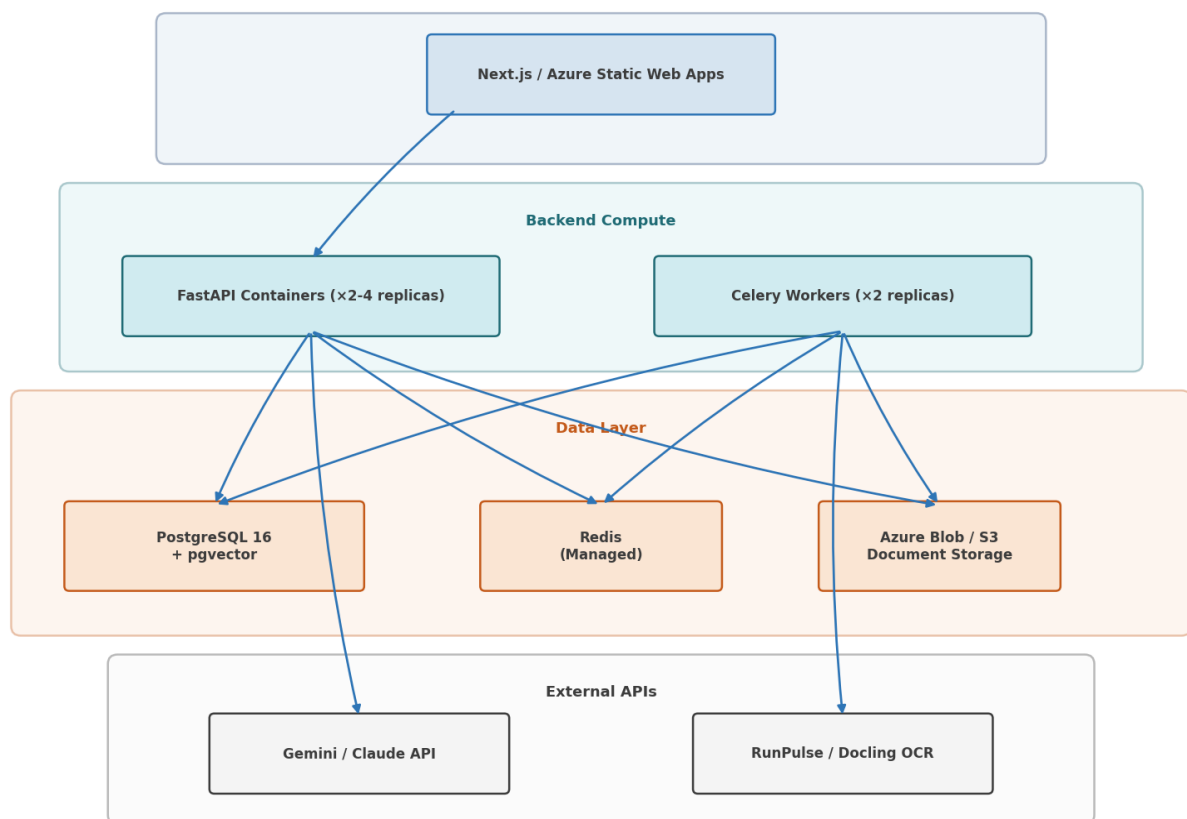
Production target architecture:

Figure 2: Production target architecture with subgraph groupings

Migration path:

1. Export Railway Postgres → SQL dump
2. Provision managed Postgres (Azure Flexible Server or AWS RDS) with pgvector extension
3. Restore dump + run Alembic migrations
4. Switch blob storage from Railway volume → Azure Blob / S3
5. Update environment variables and redeploy

2.4 Security & VPN Pipeline

To access QCI's local document server safely, a site-to-site VPN with read-only access is established:

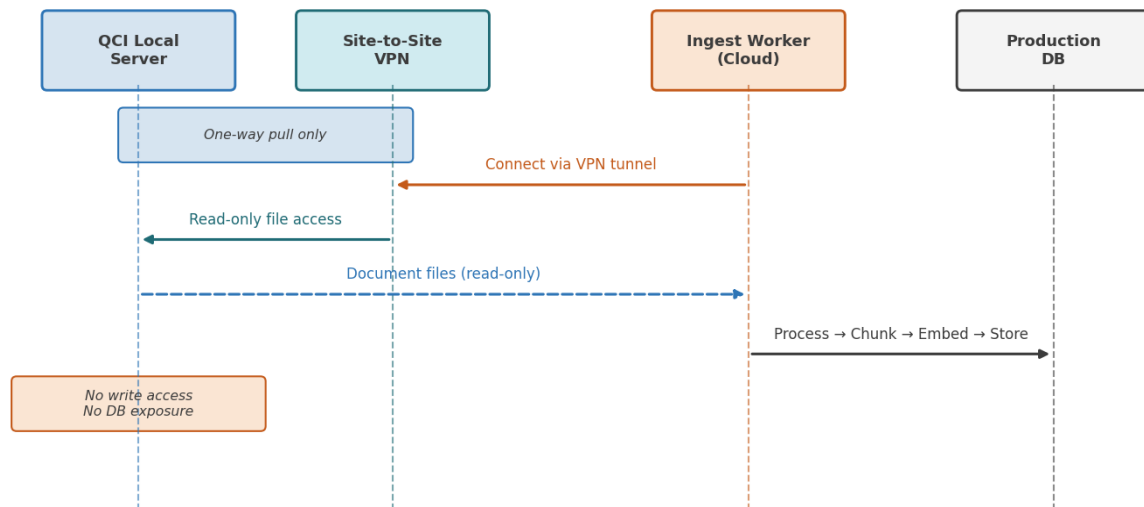


Figure 3: VPN security pipeline — read-only ingest, no write-back

Key security principles:

- Read-only mount — ingest worker can only pull files, never write back
- No direct DB exposure — QCI's repository server is never exposed to the internet
- VPN with mutual TLS — Azure VPN Gateway or WireGuard tunnel
- Audit logging — every file access logged with timestamp and operator
- QCI IT sign-off required before any VPN configuration

2.5 Model / API Choices for Document Generation

Provider	Model	Input \$/MTok	Output \$/MTok	Quality	Notes
Google (current)	Gemini Flash	\$0.30–0.50	\$1.50–3.00	Good	Free tier, already integrated
Google	Gemini Pro	\$1.25	\$5.00	Very Good	Better structure adherence
Anthropic	Claude Haiku 4.5	\$1.00	\$5.00	Very Good	Excellent structured docs
Anthropic	Claude Sonnet 4.6	\$3.00	\$15.00	Excellent	Best for complex proposals
OpenAI	GPT-4.1 mini	\$0.40	\$1.60	Good	Cheapest high-quality option
OpenAI	GPT-4.1	\$2.00	\$8.00	Very Good	Strong structured output
Open Source	Llama/Mistral/Google Gemma Models	GPU cost only	GPU cost only	Variable	Requires GPU infra, finetuning expertise

Monthly generation cost estimate (200 documents/month, ~4K tokens in + 4K out each):

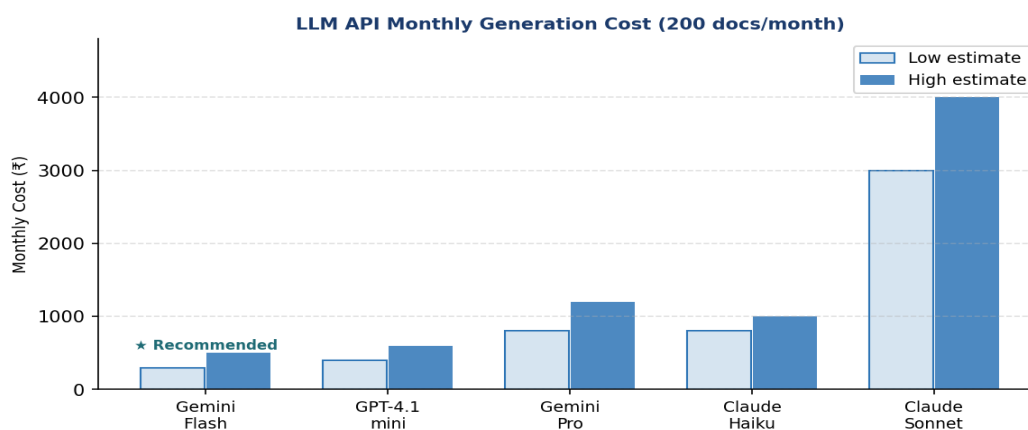


Figure 4: Estimated monthly LLM API generation costs by model

Embedding costs (one-time for 80 GB corpus + ongoing):

- Gemini embedding-001: ~₹2,000–4,000 for full corpus ingestion
- Ongoing: negligible (<₹500/month)

✓ RECOMMENDATION

Can keep Gemini Flash for generation (cheapest, already integrated, good quality). The generation costs are surprisingly low — even at scale, LLM API costs will be <₹5K/month. Can be fine-tuned easily with prompts and can always be switched for some other better option. Since quality of OCR is the main determinant for project success, there is a strong argument for using costs at the OCR stage than at Models for document generation. Self-hosting an LLM is not justified at this volume.

3. Implementation Options

Option A — Lean Production (Railway + Incremental Hardening)

"Ship fast, harden later"

Approach: Stay on Railway, upgrade to Enterprise plan, add proper auth and monitoring.

Aspect	Detail
Hosting	Railway Pro (\$20/mo base + usage) – currently being used; need to switch to enterprise plan
Database	Railway Postgres (enable backups)
OCR	Docling (bulk) + RunPulse (new docs)
LLM	Gemini Flash (current)
Auth	Add proper JWT with user accounts (no SSO yet)
VPN	WireGuard tunnel to QCI server
Frontend	Vercel free tier or Railway

Budget:

Category	Best Case	Normal Case	Worst Case
Railway hosting	₹5K/mo	₹12K/mo	₹25K/mo
RunPulse (ongoing) (on the highest sides) ~10k pages are free	₹15K/mo	₹40K/mo	₹80K/mo
Gemini API	₹500/mo	₹2K/mo	₹5K/mo
Claude Pro (dev)	₹8.5K/mo	₹8.5K/mo	₹17K/mo
Domain + SSL	₹500/mo	₹500/mo	₹500/mo
Monthly Total	₹30K	₹63K	₹1.3L
One-time setup	₹50K	₹1L	₹2L
Year 1 Total	₹4.1L	₹8.6L	₹17.6L

Timeline: 4–6 weeks · Staffing: Prabhav + QCI IT (part-time)

Trade-offs:

- ☒ Fastest to production
- ☒ Lowest cost
- ☒ Buildable with Claude Pro coding access
- ⚠ Data hosted in US (Railway) — no India data residency
- ⚠ Limited enterprise compliance features
- ✗ May not pass QCI IT security audit

Option B — Hybrid: Azure India + Managed Services

★ Recommended Option

"Government-ready without over-engineering"

Approach: Move compute and DB to Azure Central India for data residency. Use managed services to minimise ops burden.

Aspect	Detail
Hosting	Azure App Service (B2 plan, 2 instances)
Database	Azure Database for PostgreSQL Flexible Server + pgvector
OCR	Docling (bulk ingest) + Azure Document Intelligence (ongoing)
LLM	Gemini Flash via API (or Azure OpenAI for compliance)
Auth	Azure AD B2C / QCI's existing SSO
VPN	Azure VPN Gateway → QCI local server
Storage	Azure Blob Storage
Frontend	Azure Static Web Apps or Vercel

Budget:

Category	Best Case	Normal Case	Worst Case
Azure Compute (App Service)	₹8K/mo	₹18K/mo	₹35K/mo

Category	Best Case	Normal Case	Worst Case
Azure PostgreSQL	₹5K/mo	₹10K/mo	₹20K/mo
Azure Blob Storage	₹500/mo	₹1.5K/mo	₹3K/mo
Azure VPN Gateway	₹3K/mo	₹3K/mo	₹5K/mo
OCR (Azure Doc Intel)	₹2K/mo	₹5K/mo	₹15K/mo
LLM API (Gemini/OpenAI)	₹500/mo	₹3K/mo	₹8K/mo
Claude Pro (dev)	₹8.5K/mo	₹8.5K/mo	₹17K/mo
Monitoring + misc.	₹1K/mo	₹2K/mo	₹5K/mo
Monthly Total	₹29K	₹51K	₹1.1L
One-time setup	₹1.5L	₹3L	₹5L
Year 1 Total	₹5L	₹9.1L	₹18.2L

Timeline: 8–12 weeks · Staffing: Prabhav (active) + QCI IT (lead) + 1 Azure consultant (2–3 weeks, part-time, in case needed)

Trade-offs:

- ✓ Data residency in India
- ✓ Government compliance ready
- ✓ Managed services = low maintenance overhead
- ✓ Azure AD integration for SSO
- ✓ Feasible with Claude Pro coding + QCI IT support
- ⚠ More setup complexity than Railway
- ⚠ Slightly higher base cost
- ⚠ Can't manage DARPG work along with this option – need to work on this standalone to meet timelines

Option C — Full Enterprise (Dedicated Infrastructure + Ops Team)

"Maximum control, maximum cost"

Approach: Full cloud deployment with dedicated DevOps, CI/CD pipeline, staging environment, comprehensive monitoring, and self-hosted OCR.

Aspect	Detail
Hosting	Azure AKS (Kubernetes) or AWS ECS
Database	Azure PostgreSQL HA (zone-redundant)
OCR	Self-hosted Docling on GPU instances
LLM	Azure OpenAI (data stays in Azure)
Auth	Full SSO with RBAC, MFA
CI/CD	GitHub Actions → staging → production
Monitoring	Azure Monitor + Application Insights

Budget:

Category	Best Case	Normal Case	Worst Case
Azure Infrastructure	₹40K/mo	₹80K/mo	₹1.5L/mo
DevOps engineer (contract)	₹80K/mo	₹1.2L/mo	₹1.5L/mo
GPU compute (OCR)	₹15K/mo	₹30K/mo	₹50K/mo
LLM API	₹3K/mo	₹8K/mo	₹20K/mo
Monitoring + security	₹5K/mo	₹10K/mo	₹20K/mo
Monthly Total	₹1.4L	₹2.5L	₹3.9L
One-time setup	₹5L	₹10L	₹15L
Year 1 Total	₹22L	₹40L	₹62L

Timeline: 16–24 weeks · Staffing: Prabhav + dedicated DevOps engineer + QCI IT (active) + security auditor

Trade-offs:

- ☒ Maximum security, compliance, and control
- ☒ Full HA and disaster recovery
- ☒ Production-grade CI/CD
- ☒ Highest cost by far
- ☒ Requires dedicated DevOps hire
- ☒ Over-engineered for current user base
- ☒ Long timeline

4. Staffing Assessment — Can We (Prabhav + IT team) Build This Solo?

"We made the demo in a day — if we get Claude Pro coding access, can we build this standalone?"

Capability	We + Claude Pro	QCI IT Team Needed?	External Help Needed?
Backend code (FastAPI, RAG, LLM)	☒ Fully capable	No	No
Frontend (Next.js, UI)	☒ Fully capable	No	No
Database migrations, schema	☒ Fully capable	No	No
OCR pipeline (Docling/RunPulse)	☒ Proven experience	No	No
Azure/AWS initial setup	☐ Can learn, but slow	Yes — for org accounts	Maybe 1–2 days consult
VPN to QCI server	☒ Needs QCI network	Yes — mandatory	No
SSO / Azure AD integration	☐ Possible w/ guidance	Yes — QCI AD admin	No

Capability	We + Claude Pro	QCI IT Team Needed?	External Help Needed?
Security audit	✗ Cannot self-audit	Yes — mandatory	Maybe external auditor
DNS, SSL, domain	☑ Straightforward	Yes — QCI domain	No

The Practical Answer

Option A (Railway):

Yes — we can build this almost entirely solo with Claude Pro. QCI IT Team is needed only for VPN setup and domain pointing. Timeline: 4–6 weeks part-time.

Option B (Azure — recommended):

We can do ~80% solo with Claude Pro. Will need QCI IT for Azure subscription setup, VPN configuration, AD integration, and security review. A 2–3 weeks part-time Azure consultant (~₹50K–1L) would **significantly accelerate** the Azure-specific setup. Timeline: 8–12 weeks.

Option C (Enterprise):

Not feasible solo. Requires a dedicated DevOps engineer at minimum.

Claude Pro — Investment Analysis

Plan	Monthly Cost	Justification
Claude Pro (\$20/mo)	₹1,700/mo	Sufficient for moderate coding sessions
Claude Max 5× (\$100/mo)	₹8,500/mo	★ Recommended — heavy agentic coding during buildout
Claude Max 20× (\$200/mo)	₹17,000/mo	Only if hitting rate limits frequently

✓ RECOMMENDATION

ROI calculation: Claude Max at ₹8.5K/mo replaces approximately ₹2–4L/mo of additional developer hiring. Even for a 3-month build period, that is ₹25K vs ₹6–12L. The ROI is overwhelming — this is the single highest-leverage investment in the project. Can start with Claude Pro and transition to Max if needed also.

5. Timeline Comparison

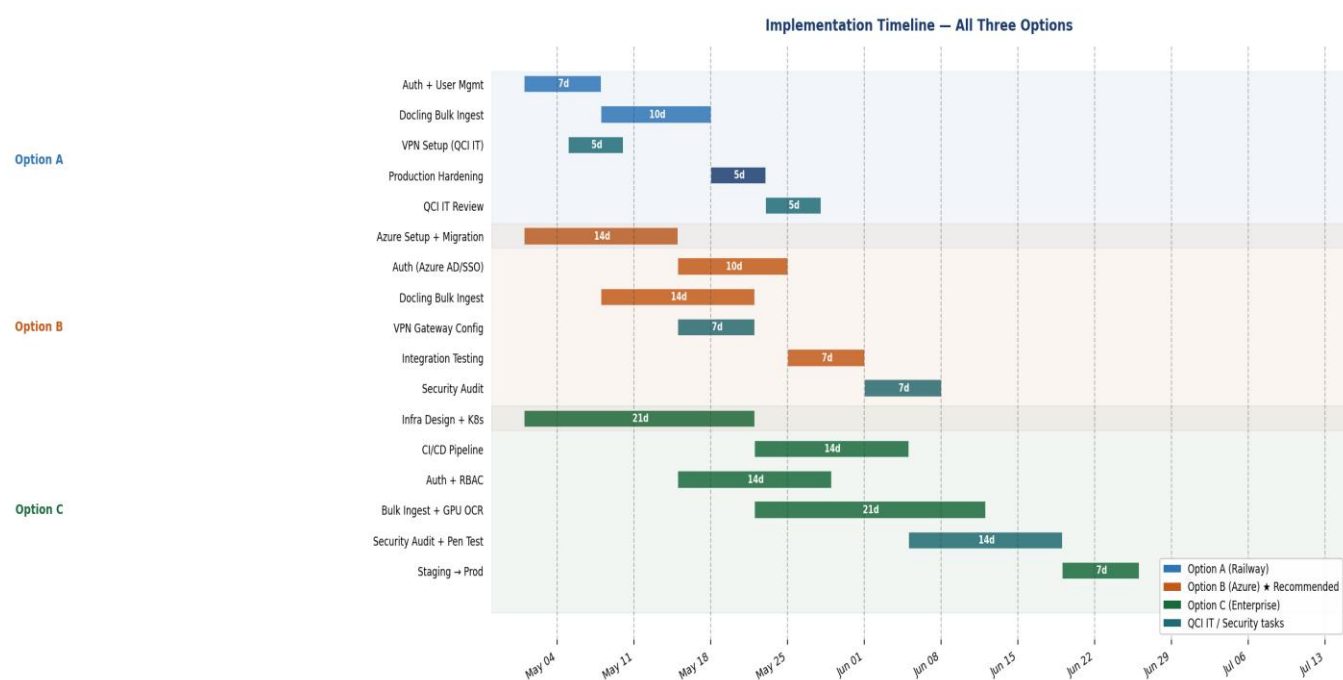


Figure 5: Implementation Gantt chart — all three options

Scenario	Option A (Railway)	Option B (Azure) ★	Option C (Enterprise)
Best case	4 weeks	8 weeks	16 weeks
Normal case	6 weeks	10 weeks	20 weeks
Worst case	8 weeks	14 weeks	24 weeks

6. Monthly Cost Comparison

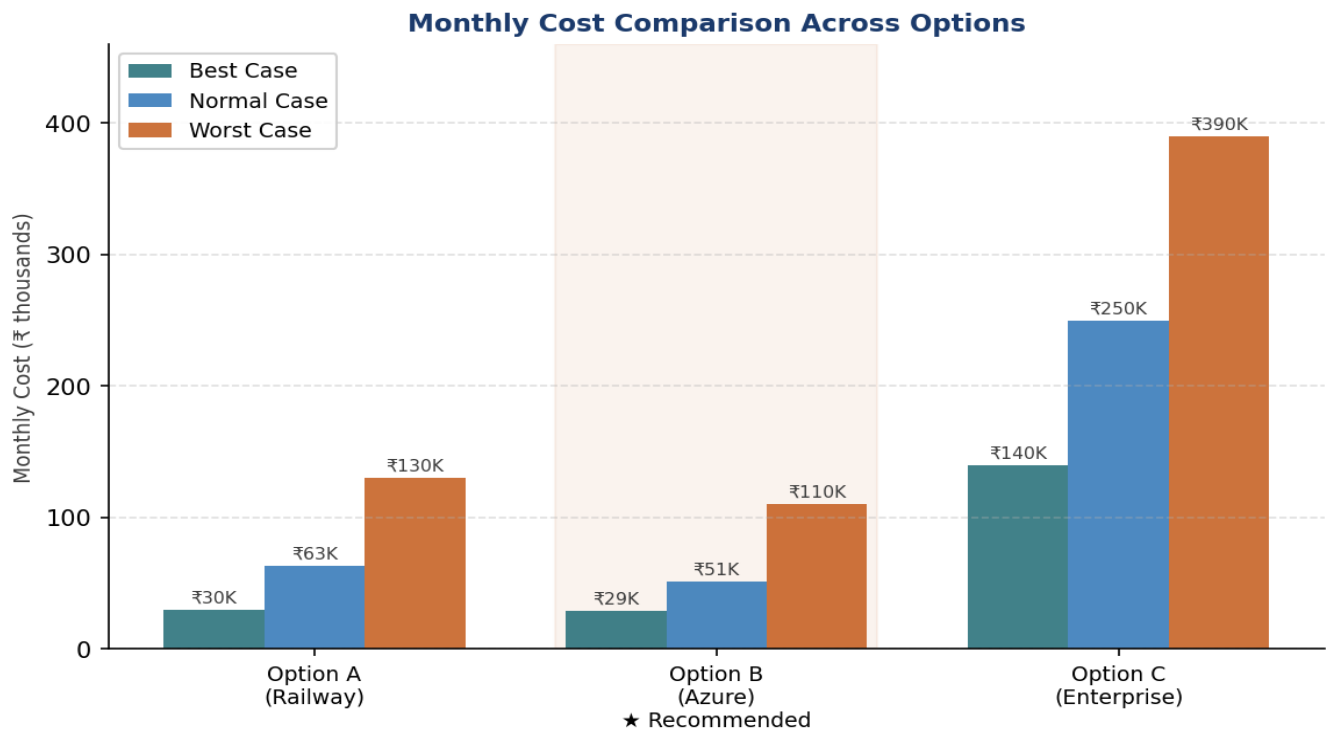


Figure 6: Monthly infrastructure cost across all three options (₹ thousands)

6.1 Maintenance & Ongoing Costs

Item	Monthly Estimate	Notes
Hosting infrastructure	₹10–50K	Depends on option chosen
LLM API usage	₹1–5K	Scales with generation volume
OCR for new documents	₹2–15K	Depends on ingest frequency
Claude Pro (dev tool)	₹8.5K	Can downgrade to Pro post-launch
Security patching + updates	4–8 hrs/month	Prabhav's time or QCI IT
Model/prompt tuning	2–4 hrs/month	As document quality improves
Total ongoing	₹25K–80K/mo	Excluding time given by Prabhav

7. Risk Matrix

Risk	Probability	Impact	Mitigation
QCI IT rejects Railway (data residency)	High	Blocks Option A	Default to Option B (Azure India)
RunPulse enterprise pricing exceeds budget	Medium	Increases OCR costs	Switch to Docling + Azure Doc Intel

Risk	Probability	Impact	Mitigation
Gemini free tier rate limits at scale	Medium	Degrades UX	Upgrade to paid tier or add caching
VPN setup delays (QCI network team)	High	Delays ingest	Start with manual upload as interim
80 GB ingest takes longer than expected	Medium	Delays go-live	Batch processing, run overnight
Key person risk (sole developer)	High	Project stalls	Document everything; use Claude for knowledge capture

8. Recommendation

⚠ IMPORTANT

Can Proceed with Option B (Azure Hybrid) with the specific action plan below. Total Year 1 Budget: ₹9–10L (normal case). Staffing: Prabhav (primary) + QCI IT (collaboration) + optional Azure consultant (₹50K–1L one-time).

Action Plan

6. Immediately — Get Claude Max subscription (₹8.5K/mo): force multiplier
7. Weeks 1–2 — Set up Azure subscription with QCI IT; provision PostgreSQL + App Service in Central India
8. Weeks 2–3 — Migrate backend from Railway to Azure App Service (Docker container, same codebase)
9. Weeks 2–4 — Run Docling bulk ingest of 80 GB corpus (can run in parallel on local machines or Azure VM)
10. Weeks 3–4 — Configure Azure VPN Gateway with QCI IT for ongoing document sync
11. Weeks 4–6 — Implement proper auth (Azure AD B2C or simple user management)
12. Weeks 6–8 — Integration testing and QCI IT security review
13. Weeks 8–10 — Soft launch with pilot group → full rollout

This option gives QCI data residency in India, government-grade compliance, and manageable costs — while remaining buildable primarily by Prabhav with AI-assisted development. It is the pragmatic middle ground between "ship fast" and "enterprise perfection".

Next step: Present to leadership for budget approval and option selection.